# QUADRATIC CHABAUTY THROUGH A HIDDEN INVOLUTION: THE PRIMITIVE SOLUTIONS OF $x^4 + 2y^3 + z^3 = 0$

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ABSTRACT. We determine all primitive integer solutions of $x^4 + 2y^3 + z^3 = 0$: they are $(\pm 1, 0, -1)$, $(\pm 1, -1, 1)$ and $(\pm 43, -203, 237)$. This was the smallest generalised Fermat equation left uninvestigated ($H = 40$) in the survey of Ratcliffe and Grechuk. A descent over $\mathbb{Q}(\sqrt[3]{2})$ reduces the equation to rational points on six genus-2 curves, four of which are settled by two-cover descent and by a classical Chabauty argument at an inert prime. The remaining pair, $B^-\colon y^2 = -C_3(x)$ and $C^+\colon y^2 = 2C_3(x)$ with $C_3(x) = 17x^6 - 72x^5 + 90x^4 + 40x^3 - 180x^2 + 144x - 40$, have Jacobians of Mordell–Weil rank 2, equal to the genus, and Néron–Severi rank 1 over $\mathbb{Q}$ and over $\mathbb{Q}(\sqrt{-2})$; neither classical Chabauty nor quadratic Chabauty over those fields applies.

We show that these curves are $\mathbb{Q}$-twists of the Bolza curve and carry an involution $\tau$ defined over a field $L$ of degree 12, whose Néron–Severi class is the unique class on which complex conjugation acts by $-1$ — exactly the class counted by the term $\operatorname{rk}\mathrm{NS}(J_{\overline{\mathbb{Q}}})^{c=-1}$ in the finiteness criterion of Balakrishnan–Dogra. The bielliptic quadratic Chabauty function of Balakrishnan–Besser–Bianchi–Müller over $L$, formed with the unique idele class character lying in the matching Galois-isotypic component, restricts to a locally analytic function on $X(\mathbb{Q}_p)$ that takes a single value on $X(\mathbb{Q})$. Computing its zeros at $p = 499$, and independently at $p = 1459$, and applying a Mordell–Weil sieve leaves exactly the four known rational points on each curve.

*Provenance.* This work was carried out in a human–AI collaboration with Claude (Anthropic): the design of the computations, the scripts and this write-up were all produced in that collaboration. Every assertion below is backed by a script and a log in the public repository, and the record of how the argument developed, including the errors found and corrected along the way, is kept there as well. The result has not been refereed. Credits and status in full: §1.5.

## 1. INTRODUCTION

### 1.1. The result.

**Theorem 1.1.** *The integer solutions of*

$$x^4 + 2y^3 + z^3 = 0, \qquad \gcd(x, y, z) = 1, \tag{1.1}$$

*are exactly*

$$(x, y, z) = (\pm 1,\ 0,\ -1), \qquad (\pm 1,\ -1,\ 1), \qquad (\pm 43,\ -203,\ 237).$$

Equation (1.1) is a generalised Fermat equation of signature $(4, 3, 3)$ and height $H = 40$. In the survey of Ratcliffe and Grechuk [RG] it is the smallest equation recorded as uninvestigated. Villines [Vi] reduced it, with Magma, to rational points on six curves of genus 2 and settled four of them, leaving open the two whose Jacobians have Mordell–Weil rank equal to the

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genus. The present paper closes that gap, and re-proves the remaining parts independently; all computations here run on PARI/GP alone.

The proof has four parts.

(A) A descent over $K = \mathbb{Q}(\sqrt[3]{2})$ reducing (1.1) to rational points on six genus-2 curves (§2).
(B) The four accessible curves: three have empty fake 2-Selmer set, and the fourth has rank 1 and is settled by Chabauty at an inert prime together with a Mordell–Weil sieve (§3).
(C) The hard pair $B^-$, $C^+$, where $\operatorname{rk} J(\mathbb{Q}) = 2 = g$, by quadratic Chabauty (§§4–6). This is the new contribution.
(D) The rank bounds used in (C), by a 2-descent over $\mathbb{Q}$ (§7).

1.2. **The obstruction, and the idea.** Write $X$ for either curve of the hard pair and $J = \mathrm{Jac}(X)$. Classical Chabauty–Coleman needs $\operatorname{rk} J(\mathbb{Q}) < g$, and here $\operatorname{rk} J(\mathbb{Q}) = 2 = g$. Quadratic Chabauty in the form of Balakrishnan–Dogra [BD] needs a supply of Néron–Severi classes: the depth-2 Chabauty–Kim set $X(\mathbb{Q}_p)_2$ is finite as soon as

$$\operatorname{rk} J(\mathbb{Q}) \;<\; g - 1 + \operatorname{rk} \mathrm{NS}(J) + \operatorname{rk} \mathrm{NS}(J_{\overline{\mathbb{Q}}})^{c=-1}, \tag{1.2}$$

$c$ denoting complex conjugation. We show (Proposition 4.3) that $\mathrm{NS}(J_{\overline{\mathbb{Q}}}) \otimes \mathbb{Q}$ has rank 4 and, as a representation of $\mathrm{Gal}(M/\mathbb{Q}) \cong \mathrm{S}_4 \times \mathbb{Z}/2$ for an explicit field $M$ of degree 48, is

$$\mathrm{NS}(J_{\overline{\mathbb{Q}}}) \otimes \mathbb{Q} \;\cong\; \mathbf{1} \oplus (\mathrm{sgn}_F \otimes \mathrm{sgn} \otimes \mathrm{std}),$$

so that $\operatorname{rk} \mathrm{NS}(J) = 1$ and complex conjugation has eigenvalues $(1, 1, 1, -1)$. The right-hand side of (1.2) is therefore $2 - 1 + 1 + 1 = 3 > 2$: finiteness holds, but the only usable class is the single anti-invariant one, and it is not defined over $\mathbb{Q}$. The same computation shows $\rho(J/F) = 1$ for $F = \mathbb{Q}(\sqrt{-2})$, so quadratic Chabauty over $F$ — the route suggested in [Vi] — has no class to use either.

The anti-invariant class turns out to be visible. The Möbius transformations permuting the six roots of $C_3$ form $\mathrm{S}_4$, so $\mathrm{Aut}(X_{\overline{\mathbb{Q}}}) \cong \mathrm{GL}_2(\mathbb{F}_3)$ has order 48 and $X$ is a $\mathbb{Q}$-twist of the Bolza curve $y^2 = x^5 - x$. Six of the nine involutory Möbius maps lift to genuine involutions $\tau$ of $X$, and these are defined over a field

$$L = \mathbb{Q}\big(\sqrt[3]{2},\ \sqrt{\sqrt[3]{2} - 1},\ \sqrt{\kappa}\big), \qquad [L : \mathbb{Q}] = 12,$$

whose degree is exactly what the Néron–Severi character predicts (the stabiliser of the anti-invariant class has order 4 in a group of order 48). Over $L$ the curve is bielliptic, with quotients $E_1, E_2$ of $j$-invariant 8000.

This puts us in the situation of the explicit bielliptic quadratic Chabauty of Balakrishnan–Besser–Bianchi–Müller [BBBM], whose function is built from an idele class character $\chi$ of the base field. The point is a *selection rule*: for $z \in X(\mathbb{Q}) \subset X(L)$ all Galois-conjugate places contribute the same local terms, so the identity survives on $\mathbb{Q}$-points only when the class and the character lie in the same Galois-isotypic component. The cyclotomic character lies in the trivial component and yields $0 = 0$. In the 3-dimensional component $\mathrm{sgn}_F \otimes \mathrm{sgn} \otimes \mathrm{std}$ that matches $\tau_*$ there is, we show, exactly one character $\chi_Z$ up to scaling (Proposition 5.3); and it vanishes on the uniformisers of every prime of $L$ above 2 and 3, which is what makes the resulting identity independent of the reduction types at the bad primes.

The outcome (Theorem 5.5) is that

$$\rho(z) \;=\; \sum_{i=1}^{12} c_i \big[\lambda_i(\varphi_1 z) - \lambda_i(\varphi_2 z)\big] \;-\; 2 \sum_{i=1}^{12} c_i \log_p \sigma_i(X(z)) \;-\; a(z)^{\mathsf{T}} G a(z)$$

is a locally analytic function on $X(\mathbb{Q}_p)$ taking the single value $-\Omega_{397}$ on $X(\mathbb{Q})$. Its zero set is computed disk by disk, and a Mordell–Weil sieve cuts the survivors down to the four known rational points on each curve.

1.3. **What is new.**

(1) An explicit quadratic Chabauty computation for a curve over $\mathbb{Q}$ whose extra Néron–Severi class is defined only over a field of degree 12, found from the automorphism group of a twist of the Bolza curve, with the field of definition matching the Galois character of $\mathrm{NS}(J_{\overline{\mathbb{Q}}})$.
(2) The identification of the character by linear algebra from the action of the involutions on holomorphic forms together with the $p$-adic logarithms of the units of $L$, with the vanishing above 2 and 3 as a by-product.
(3) $p$-adic heights at twelve places of a degree-12 field, computed through integer models congruent to the completions modulo $p^{30}$, and division-polynomial recurrences for $\psi_{514}$ in place of a polynomial of degree ≈130,000.
(4) An index-free Mordell–Weil sieve: $p$-integrality of the coefficient vector removes the saturation hypothesis usually needed.
(5) The observation that the endgame proposed in [Vi] over $\mathbb{Q}(\sqrt{-2})$ cannot work, since $\rho(J/\mathbb{Q}(\sqrt{-2})) = 1$.

1.4. **What the proof rests on.** Parts (A), (B) and (D) are self-contained exact or $p$-adic computations. Part (C) additionally imports:

(i) $p$-adic heights on elliptic curves over a number field for a non-cyclotomic idele class character: that $h^\chi$ is a quadratic form, that it decomposes into local terms, that the local term at $w \nmid p$ of good reduction is $\chi_w(\pi_w)\max(0, -v_w(x(Q)))$ on a model minimal at $w$, and that at $\mathfrak{p} \mid p$ of good ordinary reduction it is the canonical Mazur–Tate height computed from the $\sigma$-function [MT, Bi, BBBM].
(ii) The bielliptic quadratic Chabauty identity of [BBBM, Thm. 1.5] over a number field, with the normalisation of the logarithmic term as in [BP, Thm. 2.3].
(iii) The finiteness theorem of [BD, Prop. 2] (quoted in [Do]); this is used only as motivation, the finiteness of the zero set being established computationally by Strassman bounds.
(iv) Standard Coleman integration on residue disks.
(v) PARI's rigorous $p$-adic precision tracking. The only truncation is the series tail on each residue disk, controlled by the observed linear growth of coefficient valuations and confirmed at higher precision.

1.5. **Provenance and credits.** This work was done in collaboration with Claude (Anthropic): the design of the computations, the PARI/GP and Python scripts, and the present write-up were all produced in that collaboration. The author is nonetheless answerable for everything asserted here, which is what the name on the title page means. Claude is not an author and could not be one: it can neither stand behind the result nor answer for it if the result is wrong.

The author is aware that many mathematicians are uneasy about work produced with AI systems, and regards that unease as reasonable: such systems can produce confident arguments that are wrong, and checking them costs the time of people who did not ask to spend it. The author came to this problem in order to learn the mathematics around it, and used Claude as a tool for doing so. That does not make the result more likely to be right, and no one is asked to take it on trust.

The reader should treat the mathematics accordingly — that is, critically, and with the scripts and logs in hand. Everything needed to do so is public (§9): every proposition names

the script and the log that establish it (Table 1), and `RESULTS.md` in the repository is a chronological record of how the argument developed, including the several errors that were found and corrected and the checks that caught them.

This is therefore a research record, not a refereed result. The computations have been repeated at a second prime and cross-checked against the independent Magma computations of [Vi] wherever the two overlap, and they agree throughout; but the argument of §5, and in particular the precise relationship between the identity proved there and [BBBM, Thm. 1.5], has not been examined by anyone else. The questions the author would most like an expert to consider are collected in `REFEREE_NOTES.md` in the repository.

One part of the argument has been checked by machine. The elementary passage from Lemma 2.1 to the six curves, and from their rational points back to the solutions, is formalised in Lean 4 with Mathlib (§9). The formal theorem is conditional: it takes as hypotheses Lemma 2.1, Propositions 3.1 and 3.2, and Corollary 6.4, none of which is formalised. In particular the quadratic Chabauty argument, which is the new part, is not.

**Convention 1.2.** Throughout, $\iota$ denotes the hyperelliptic involution $(x, y) \mapsto (x, -y)$; $\theta = \sqrt[3]{2}$; $K = \mathbb{Q}(\theta)$; $F = \mathbb{Q}(\sqrt{-2})$; and

$$C_3(x) = 17x^6 - 72x^5 + 90x^4 + 40x^3 - 180x^2 + 144x - 40.$$

For a curve $C\colon y^2 = f(x)$ of genus 2 we write $J = \mathrm{Jac}(C)$. Every assertion labelled with a script name is established by that script; the scripts and their logs are in the repository (§9), and Table 1 indexes them.

## 2. Descent to six curves of genus two

Let $\theta = \sqrt[3]{2}$ and $K = \mathbb{Q}(\theta)$. Then $K$ has class number 1 (certified with `bnfcertify`), fundamental unit $\varepsilon = \theta - 1$ of norm 1, and 2 and 3 are totally ramified.

**Lemma 2.1.** *Let $(x, y, z)$ be a primitive solution of (1.1). Then $\gcd(y, z) = 1$ and $\alpha := z + y\theta$ satisfies $\mathrm{N}_{K/\mathbb{Q}}(\alpha) = z^3 + 2y^3 = -x^4$. The three conjugates of $\alpha$ are pairwise coprime away from the primes above 2 and 3, and primitivity forces the common factors there to be trivial; hence $(\alpha) = \mathfrak{a}^4$ for an ideal $\mathfrak{a}$. Since $h_K = 1$ and $\mathcal{O}_K^\times = \langle -1, \varepsilon \rangle$ we may write*

$$\alpha = -\varepsilon^k \gamma^4, \qquad \gamma \in \mathbb{Z}[\theta], \quad k \in \{0, 1, 2, 3\},$$

*the sign being normalised by replacing $x$ with $-x$ if necessary.*

*Proof.* The ideal argument is the standard one for norm forms of cubic fields; see `descent_explore.py`. □

**Lemma 2.2.** *Write $\gamma = a + b\theta + c\theta^2$. The coefficient of $\theta^2$ in $-\varepsilon^k\gamma^4$ must vanish, which is a smooth plane quartic $C_k(a, b, c) = 0$ over $\mathbb{Q}$, of genus 3. Primitivity forces $\gamma$ to be a unit at the primes above 2 and 3, that is, $a$ odd and $a + 2b + c \not\equiv 0 \pmod 3$. The curve $C_3$ has no 2-adic points, and $C_2$ has no 2-adic point with $a$ odd. Hence every primitive solution comes from a rational point of $C_0$ or $C_1$ satisfying the unit conditions, with $x = \mathrm{N}(\gamma)$ and $z + y\theta = -\varepsilon^k\gamma^4$.*

*Proof.* `descent_explore.py`, `diagonal.py`; the explicit quartics are recorded in `RESULTS.md`. □

**Lemma 2.3** (Étale double covers and the twist set)**.** *Put $\delta = \gamma^2$. The condition $[\theta^2](-\varepsilon^k\delta^2) = 0$ is a smooth conic $Q_k(d_0, d_1, d_2) = 0$ in the coordinates of $\delta = d_0 + d_1\theta + d_2\theta^2$ (Gram*

*determinant* 1 *for both* $k$*), with the rational point* $(1:0:0)$*. Parametrising by the lines through that point,*

$$\begin{aligned} k=0:\quad & Q_0 = -d_1^2 - 2d_0d_2, \qquad \delta_0(s,t) = (s^2,\, -2st,\, -2t^2),\\ & f_0(s,t) = \mathrm{N}(\delta_0) = s^6 - 40s^3t^3 - 32t^6;\\ k=1:\quad & Q_1 = d_1^2 - 2d_0d_1 + 2d_0d_2 - 2d_2^2, \qquad \delta_1(s,t) = (2t^2 - s^2,\, 2s(t-s),\, 2t(t-s)),\\ & f_1(s,t) = -t^6 C_3(s/t). \end{aligned}$$

*Every rational point of the conic is* $\delta_k(s:t)$ *for a unique* $(s:t) \in \mathbb{P}^1(\mathbb{Q})$*; with* $s,t$ *coprime integers,* $\delta = \mu\cdot\delta_k(s,t)$ *for a unique* $\mu \in \mathbb{Q}^\times$*, and taking norms,* $\mu f_k(s,t) = (\mathrm{N}(\gamma)/\mu)^2$*. Then* $\mu \in \{\pm1, \pm\frac12\}$*, and* $\mu = \pm1$ *when* $k = 0$*. Consequently every primitive solution gives a rational point on one of the six curves*

$$A^{\pm}\colon y^2 = \pm\big(x^6 - 40x^3 - 32\big), \tag{2.1}$$

$$B^{\mp}\colon y^2 = \mp C_3(x), \qquad C^{\mp}\colon y^2 = \mp 2\, C_3(x). \tag{2.2}$$

*Proof.* Each point of $C_k(\mathbb{Q})$ gives the rational point $\big(s/t,\ \mathrm{N}(\gamma)/(\mu t^3)\big)$ on $H_\mu\colon w^2 = \mu f_k(x,1)$, and the point of $C_k$ is recovered from it by $\delta = \mu\delta_k(s,t)$, $\gamma = \sqrt{\delta}$. It remains to pin down the twist class of $\mu$.

*(i) Content.* For coprime $(s,t)$ the content of $\delta_k(s,t)$ is 1 if $s$ is odd and 2 if $s$ is even. Indeed an odd prime dividing all three coordinates would divide both $s$ and $t$: for $k=0$ it divides $s^2$ and $2t^2$; for $k=1$ it divides $2s(t-s)$ and $2t(t-s)$, hence $t-s$, hence $2t^2 - s^2 \equiv t^2$. The claim modulo 4 follows by inspecting the coordinates $(s^2, 2st, 2t^2)$, resp. $(2t^2 - s^2, \dots)$ with the other two even.

*(ii)* $\gamma^2$ *is primitive.* $\gamma$ is primitive; if $p$ divided all coordinates of $\gamma^2$ then $\gamma$ would be nilpotent in $\mathbb{Z}[\theta]/(p)$, which is reduced for $p \nmid 6$, while for $p = 2, 3$ the nilpotents are $(\theta)$ and $(\theta+1)$, excluded by the unit conditions of Lemma 2.2. Hence $\mu = \pm1/\operatorname{cont}(\delta_k(s,t)) \in \{\pm1, \pm\frac12\}$.

*(iii) The case* $k = 0$. The $\theta^0$-coordinate of $\delta = \gamma^2$ is $a^2 + 4bc$, which is odd, while that of $\mu\delta_0(s,t)$ is $\mu s^2$; so $s$ is odd, the content is 1, and $\mu = \pm1$. For $k = 1$ both contents occur. Multiplying $w^2 = \pm\frac12 f_1$ by 4 gives the models (2.2). □

These are the six curves of [Vi]. The known rational points transport as follows (`etale.py`): the points $(0,1,0)$, $(1,0,-1)$, $(1,0,0)$ of $C_0$ give $(0:1)$, $(1:-1)$, $(1:0)$ on $A^+$; and $(1,-2,0)$, $(1,0,0)$, $(1,-1,2)$, $(1,1,0)$ of $C_1$ give $(-1,\pm15)$, $(1,\pm1)$ on $B^-$ and $(6/5,\pm172/125)$, $(2,\pm12)$ on $C^+$. The point $(6/5,\pm172/125)$ is the solution $(\pm43, -203, 237)$.

**Remark 2.4.** The sign of $\mu$ is the sign of $f_k(s,t)$. The point of $C_0$ with $\gamma = \theta$, i.e. $\delta = \theta^2$ and $(s,t) = (0,1)$, has $s$ even and is excluded by the unit condition "$a$ odd"; it corresponds to an imprimitive solution. So the content-2 chart for $k=0$, which would land on the twists $y^2 = \pm2(x^6 - 40x^3 - 32)$, never arises from a primitive solution.

## 3. The four accessible curves

**Proposition 3.1.** $A^-(\mathbb{Q}) = B^+(\mathbb{Q}) = C^-(\mathbb{Q}) = \emptyset$.

*Proof.* Let $C\colon y^2 = f(x)$ be one of the three, with leading coefficient $f_6 \in \{-1, 17, -34\}$ respectively, let $L_6 = \mathbb{Q}[x]/(f)$ — so $L_6 \cong \mathbb{Q}[u]/(u^6 - 3u^4 - 2u^3 + 3u^2 + 6u + 1)$ for $A^\pm$ and $\mathbb{Q}[u]/(u^6 - 3u^4 + 3u^2 - 3)$ for $B^\pm, C^\pm$, both of class number 1 (certified by `bnfcertify`) — let $\vartheta$ be the image of $x$, and let $S$ be the set of primes dividing $2\operatorname{disc}(f)f_6$, namely $\{2,3\}$ resp. $\{2,3,17\}$.

Since $f$ has no rational root and $f_6$ is not a square, $C$ has no Weierstrass and no rational point at infinity. For an affine point $(x,y)\in C(\mathbb{Q})$,

$$\mathrm{N}_{L_6/\mathbb{Q}}(x-\vartheta)=f(x)/f_6=y^2/f_6\equiv f_6 \pmod{\mathbb{Q}^{\times 2}},$$

and the class of $x-\vartheta$ in $L_6^\times/L_6^{\times 2}\mathbb{Q}^\times$ is unramified outside $S$. (This is [BS, Lemma 4.3]: for $p\notin S$, $f$ is squarefree mod $p$ with unit leading coefficient, so $x$ mod $p$ is a root of exactly one — linear — factor of $f$ mod $p$; a single prime $\mathfrak{p}\mid p$ divides $x-\vartheta$, and $v_\mathfrak{p}(x-\vartheta)=v_p(y^2/f_6)$ is even; and if $v_p(x)<0$ then $(x-\vartheta)/x\equiv 1$ modulo every prime above $p$.) Multiplying by an element of $\mathbb{Q}^\times$ changes the norm by a sixth power, so a rational point would produce an $S$-unit $\delta$ of $L_6$ with $\mathrm{N}(\delta)\in f_6\mathbb{Q}^{\times 2}$.

The norms of the $S$-units modulo squares span the subgroup $\langle -2,-3\rangle$ of $\mathbb{Q}(S,2)$ for the sextic field of $A^\pm$ and $\langle -2,-3,-17\rangle$ for that of $B^\pm, C^\pm$ (`sel2/certify.gp`). Neither subgroup contains $-1$ (needed for $A^-$), 17 (for $B^+$) or $-34$ (for $C^-$) — while, as they must, they do contain $-17$ (for $B^-$) and 34 (for $C^+$). Hence $C(\mathbb{Q})=\emptyset$. □

This is the emptiness of the fake 2-Selmer set of Bruin–Stoll [BS] at their norm step, and reproduces the corresponding result of [Vi]. The script `sel2/fakeselmerset.gp` implements the full fake Selmer set — norm condition, real place, and the exact local images $\mu_p(C(\mathbb{Q}_p))$ by recursive subdivision of $p$-adic disks — and returns $W=\emptyset$ for these three curves already at the norm step.

**Proposition 3.2.** $A^+(\mathbb{Q})=\{\infty_\pm,\,(-1,\pm 3)\}$.

*Proof. (a) Rank.* $f=x^6-40x^3-32$ is irreducible, so $J(\mathbb{Q})[2]=0$, and `sel2/pipeline.gp` gives $\dim \mathrm{Sel}^2(J/\mathbb{Q})=1$ — the sextic field has the quadratic subfield $\mathbb{Q}(\sqrt{3})$, since $f=(x^3-20)^2-432$, so here $\mathrm{Sel}^2=\mathrm{Sel}^2_{\mathrm{fake}}$ — whence $\mathrm{rk}\,J(\mathbb{Q})\le 1$. The class $D=[(-1,3)-\infty_+]$ has infinite order, its images on the elliptic quotients below having infinite order. So $\mathrm{rk}\,J(\mathbb{Q})=1$. The torsion subgroup is $\mathbb{Z}/3$, generated by $T=[\infty_--\infty_+]$: the gcd of $\#J(\mathbb{F}_\ell)$ over $5\le\ell<400$ is 3, and $T$ has order 3.

*(b) Bielliptic structure.* Since $f(c/x)=-32f(x)/x^6$ for $c^3=-32$, $X=A^+$ carries the involution $\tau(x,y)=(c/x,\,4\sqrt{-2}\,y/x^3)$ over $M_0=\mathbb{Q}(\sqrt[3]{2},\sqrt{-2})=\mathbb{Q}(\sqrt[6]{-2})$, with fixed points $x_\pm=\pm w^5$ where $w^6=-2$, $\sqrt[3]{2}=-w^2$, $\sqrt{-2}=-w^3$ and $c=-2w^4$. In the coordinates $\mathcal{X}=(x-x_+)/(x-x_-)$, $\mathcal{Y}=y(1-\mathcal{X})^3$ the curve becomes $\mathcal{Y}^2=A\mathcal{X}^6+B\mathcal{X}^4+C\mathcal{X}^2+D$ with $A=-64+160w^3$, $B=-960-480w^3$, $C=-960+480w^3$, $D=-64-160w^3$ (all in $\mathbb{Q}(\sqrt{-2})$), and $\tau\iota$ is $(\mathcal{X},\mathcal{Y})\mapsto(-\mathcal{X},\mathcal{Y})$. The quotients

$$E_1\colon v^2=u^3+Bu^2+ACu+A^2D,\qquad E_2\colon v^2=u^3+Cu^2+BDu+AD^2,$$

with $\varphi_1=(A\mathcal{X}^2,A\mathcal{Y})$ and $\varphi_2=(D/\mathcal{X}^2,D\mathcal{Y}/\mathcal{X}^3)$, have $j=8000$ (CM by $\mathbb{Z}[\sqrt{-2}]$), conductor of norm 9 over $M_0$, $E_i(M_0)_{\mathrm{tors}}\cong\mathbb{Z}/6$, and $E_2$ is the quadratic twist of $E_1$ by $-2$. $J$ is isogenous to $E_1\times E_2$ over $M_0$ (`aplus/bi_Aplus.gp`).

*(c) Chabauty at $p=5$.* The prime 5 is inert in $\mathbb{Q}(\sqrt{-2})$ and has three primes of degree 2 in $M_0$; fix $\mathfrak{p}\mid 5$, so $M_{0,\mathfrak{p}}=\mathbb{Q}_{25}=\mathbb{Q}_5(\sqrt{2})$. Let $\mathcal{L}_i(z)=\mathrm{Log}_{E_i,\mathfrak{p}}(\varphi_i(z))\in\mathbb{Q}_{25}$ be the formal-group logarithms, computed from the formal group law of the $a_1=a_3=0$ model after multiplication by $\#E_i(\mathbb{F}_{25})=36$ (`aplus/chab5_Aplus.gp`). The differentials $\varphi_i^*\omega_{E_i}$ form a basis of $H^0(X_{M_0},\Omega^1)$, so for $z\in X(\mathbb{Q})$, writing $[z-\infty_+]=\lambda D+\epsilon T$ with $\lambda\in\mathbb{Q}$ and $\epsilon\in\mathbb{Z}/3$ (and $\mathrm{Log}\,T=0$),

$$(\mathcal{L}_1(z),\mathcal{L}_2(z))=\lambda\cdot(\mathcal{L}_1(D),\mathcal{L}_2(D)),\qquad \lambda\in\mathbb{Q}_5,$$

where $\mathcal{L}_i(D):=\mathcal{L}_i((-1,3))$. At $(-1,\pm 3)$ this holds with $\lambda=\pm 1$ exactly, to $O(5^{44})$, and $\rho(z):=\mathcal{L}_2(D)\mathcal{L}_1(z)-\mathcal{L}_1(D)\mathcal{L}_2(z)$ vanishes at all four known points. On each of the six residue disks of $X(\mathbb{Q}_5)$ — $x\equiv 1,4 \bmod 5$ with two signs of $y$, and the two disks at infinity —

expand $\mathcal{L}_i$ by termwise integration of $\varphi_i^*\omega_{E_i}$ and solve $\rho = 0$. Because $E_2 \cong E_1$ over $\mathbb{Q}_{25}$ the $\mathbb{Q}_5$-component of $\rho$ vanishes identically and the condition is the $\sqrt{2}$-component; its Strassman indices are $1, 1, 2, 2, 1, 1$, giving exactly 8 zeros in $X(\mathbb{Q}_5)$:

- $\infty_\pm$ and $(-1, \pm 3)$, with $\lambda = 0, 0, \pm 1$;
- $(-2\sqrt[3]{2}, \pm 12\sqrt{6})$ with $\lambda = 0$, a torsion class of order 3 — verified exactly over $\mathbb{Q}(\sqrt[6]{-2}, \sqrt{-3})$ (`aplus/tors12.gp`), so no rational point lies in its disk, that disk having a unique zero;
- one pair $z_1, \iota z_1$ in the disk $x \equiv 1 \pmod 5$ with $\lambda \equiv \pm 8316 \pmod{5^6}$, so $\lambda \not\equiv 0, \pm 1 \pmod{25}$.

*(d) Sieve.* A rational point $z$ in the disk $x \equiv 1$ would have $\lambda(z) = n/m \in \mathbb{Q}$ with $5 \nmid m$: the logarithms of $E_i(\mathbb{Q}_{25})$ have valuation $\geq 1$ (formal group, and $\#E_i(\mathbb{F}_{25}) = 36$ is a 5-unit) while $\mathcal{L}_i(D)$ has valuation exactly 1, so if $D = mG$ for a generator $G$ then $5 \nmid m$. Hence for every prime $\mathfrak{q}$ of $M_0$ with $5^k \mid \#E_i(\mathbb{F}_\mathfrak{q})$ the 5-primary parts satisfy $\pi_i(\varphi_i(\bar{z}) - \varphi_i(\infty_+)) = \lambda(z)\pi_i(\varphi_i(\bar{D}))$ in $E_i(\mathbb{F}_\mathfrak{q})$ (the class $T$ has order 3), with $\lambda(z) \equiv \lambda(z_1) \pmod{5^6}$ if $z$ lies in the disk of $z_1$. The script `aplus/sieve5_Aplus.gp` enumerates $A^+(\mathbb{F}_q)$ and collects the residues $\lambda \bmod 5^4$ realised: at the three degree-2 primes above $q = 149$ (where $\#E_i(\mathbb{F}_\mathfrak{q}) = 22500 = 2^2 3^2 5^4$) 175 of the 625 residues occur, and at a degree-2 prime above $q = 599$ (where $\#E_i = 360000 = 2^6 3^2 5^4$) 275 of 625. The residues $0, \pm 1$ of the known points occur at both, while $\lambda(z_1) \equiv 191$ and $\lambda(\iota z_1) \equiv 434 \pmod{625}$ occur at neither $q = 599$. So the disk $x \equiv 1$ contains no rational point. The disks $x \equiv 4$ have Strassman index 2 and contain the exhibited zeros $(-1, \pm 3)$ and $(-2\sqrt[3]{2}, \pm 12\sqrt{6})$; the disks at infinity have index 1 and contain $\infty_\pm$. □

**Remark 3.3.** At a prime split in $\mathbb{Q}(\sqrt{-2})$ — we first used $p = 43$, `aplus/linchab_Aplus.gp` — the same computation has 66 zeros: the 4 rational points, 12 algebraic points carrying torsion classes ($x^3 = 2$, $x^3 = -16$, over $\mathbb{Q}(\sqrt[6]{-2}, \sqrt{-3})$, of exact orders 3 and 6), and 50 others, of which the 43-adic sieve removes all but 14. The survivors include the $\mathbb{Q}(\sqrt{-2})$-points $(2, \pm 12\sqrt{-2})$ and $(\pm 2\sqrt{-2}, \pm(16 \pm 20\sqrt{-2}))$, whose classes have 43-adic logarithms on the line of $J(\mathbb{Q})$ because the CM action of $\mathbb{Z}[\sqrt{-2}]$ on $E_i$ commutes with the 43-adic logarithms when 43 splits in $\mathbb{Q}(\sqrt{-2})$: Chabauty at a split prime sees $J(\mathbb{Q}(\sqrt{-2}))$ rather than $J(\mathbb{Q})$. This is why the inert prime 5 is the right choice.

## 4. The hard pair: automorphisms and Néron–Severi

From here on $X$ denotes $B^-\colon y^2 = -C_3(x)$; the argument for $C^+\colon y^2 = 2C_3(x)$, its quadratic twist by $-2$, is identical with the twisted data, and all numerical values quoted are those for $B^-$ unless stated otherwise. $X$ has the rational points $P_1 = (1, 1)$, $P_2 = (-1, 15)$ and their $\iota$-images; $J = \mathrm{Jac}(X)$.

**Proposition 4.1.** *(i)* $\mathrm{Aut}(X_{\overline{\mathbb{Q}}}) \cong \mathrm{GL}_2(\mathbb{F}_3)$*, of order* 48*;* $X$ *is a* $\mathbb{Q}$*-twist of the Bolza curve* $y^2 = x^5 - x$*.*

*(ii) The Möbius maps permuting the six roots of* $C_3$ *form* $\mathrm{S}_4$*; nine of them are involutions, and the six that lift to involutions* $\tau$ *of* $X$ *are* $x \mapsto (-x + b)/(cx + 1)$ *with*

$$81b^6 - 540b^5 + 1494b^4 - 2240b^3 + 1980b^2 - 1008b + 232 = 0,$$

*the lift being* $(x, y) \mapsto \big(\mu(x),\, \kappa s\, y/(cx + 1)^3\big)$*, where* $\kappa = 1 + bc$ *and* $s^2 = \kappa$*. (The other three involutory Möbius maps, those with* $b^3 - 18b^2 + 36b - 20 = 0$*, lift to automorphisms of order* 4*.)*

*(iii) The Möbius parts are defined over* $K_6 = \mathbb{Q}(\sqrt[3]{2}, \sqrt{\sqrt[3]{2}-1})$ *and the involutions over* $L = K_6(\sqrt{\kappa})$*, a field of degree* 12 *with*

$$L = \mathbb{Q}[t]/(t^{12} + 6t^{10} + 3t^8 - 28t^6 - 33t^4 + 6t^2 - 3),$$

*of signature* $(2,5)$*, discriminant* $2^{28}3^{13}$ *and class number* 1*, in which* 2 *is totally ramified and* $3 = \mathfrak{p}_1^3\mathfrak{p}_2^6\mathfrak{p}_3^3$.

*(iv) Over* $L$*,* $X$ *is bielliptic: with* $x_\pm = (-1 \pm s)/c$ *the fixed points of* $\mu$ *and*

$$\mathcal{X} = \frac{x - x_+}{x - x_-}, \qquad \mathcal{Y} = y(\mathcal{X} - 1)^3,$$

*the curve becomes* $\mathcal{Y}^2 = F(\mathcal{X}) = A\mathcal{X}^6 + B\mathcal{X}^4 + C\mathcal{X}^2 + D$ *with* $A = f(x_-)$*,* $D = f(x_+)$*, and* $\tau(\mathcal{X}, \mathcal{Y}) = (-\mathcal{X}, \mathcal{Y})$*. The quotients*

$$E_1 \colon Y'^2 = X'^3 + BX'^2 + ACX' + A^2D, \qquad E_2 \colon Y'^2 = X'^3 + CX'^2 + BDX' + AD^2,$$

*reached by* $\varphi_1 = (A\mathcal{X}^2, A\mathcal{Y})$ *and* $\varphi_2 = (D/\mathcal{X}^2, D\mathcal{Y}/\mathcal{X}^3)$*, have* $j = 8000$ *(CM by* $\mathbb{Z}[\sqrt{-2}]$*), admit global minimal models over* $L$*, have minimal discriminants supported on the prime above* 2 *and one prime above* 3*, and* $E_i(L)_{\mathrm{tors}} \cong \mathbb{Z}/2$.

*Proof.* Exact computation. `invol.gp` finds the 24 Möbius maps numerically, and `invol4.gp`, `invol8.gp` identify them exactly by resultants: the nine involutory maps have $b$ a root of $(b^3-18b^2+36b-20)\cdot(81b^6-\cdots)\cdot(-C_3(b))$, the last factor being spurious. One has $\lambda/(1+bc)^3 = +1$ for the six genuine involutions and $-1$ for the three lifts of order 4. Fields, discriminants, class numbers, minimal models and torsion: `invol8.gp`, `step2a.gp`. The normalisation of the quotient maps follows [BP]. □

**Remark 4.2.** The natural quotient models are non-minimal at two degree-1 primes above 397, where $A$ and $D$ respectively have valuation $-6$. This is the source of the constant $\Omega_{397}$ in Theorem 5.5.

**Proposition 4.3** (Galois action on NS)**.** *Let* $N$ *be the* $\mathrm{S}_4$*-field obtained as the Galois closure of* $\mathbb{Q}(\sqrt[3]{2}, \omega, \sqrt{-1-\sqrt[3]{2}})$*, let* $F = \mathbb{Q}(\sqrt{-2})$ *and* $M = NF$*, so* $[M : \mathbb{Q}] = 48$ *and* $\mathrm{Gal}(M/\mathbb{Q}) \cong \mathrm{S}_4 \times \mathbb{Z}/2$*. Then*

$$\mathrm{NS}(J_{\overline{\mathbb{Q}}}) \otimes \mathbb{Q} \;\cong\; \mathbf{1} \oplus (\mathrm{sgn}_F \otimes \mathrm{sgn} \otimes \mathrm{std}).$$

*In particular* $\rho(J/\mathbb{Q}) = \rho(J/F) = 1$*, and complex conjugation acts with eigenvalues* $(1,1,1,-1)$*: there is exactly one anti-invariant class, namely the class* $Z = \tau_*$ *of an involution, defined over* $L$.

*Proof.* Frobenius traces (`nschar2.gp`). Every prime inert in $F$ has trace 0, so $V_\ell(J) = \mathrm{Ind}_F^{\mathbb{Q}}\psi$; at split primes the Frobenius polynomial factors over $F$ (constant terms $\pm\pi^2, \pm\bar{\pi}^2$) but over $\mathbb{Z}$ only when it is even. Hence $\mathrm{End}_F^0(J) = F$, and since the Rosati involution on an imaginary quadratic field is complex conjugation, $\rho(J/F) = 1$.

The trace of Frobenius on $\mathrm{NS}(J_{\overline{\mathbb{Q}}}) \otimes \mathbb{Q}_\ell$ — the count of root-of-unity eigenvalues among the $\alpha_i\alpha_j/p$, the 2-dimensional transcendental part contributing 0 at inert primes — depends only on the splitting of $p$ in $F$ and in $N$:

| class in $N$ | condition on $p$ | $F$-split | $F$-inert |
|---|---|---|---|
| 3-cycles | $p \equiv 1\ (3)$, $x^3 - 2$ irred. mod $p$ | 1 | 1 |
| double transp. | $p \equiv 1\ (3)$, three roots, $(-1-r \mid p) = [-1,-1,1]$ | 0 | 2 |
| identity | $p \equiv 1\ (3)$, three roots, all $(-1-r \mid p) = 1$ | 4 | $-2$ |
| transpositions | $p \equiv 2\ (3)$, $(-1-r \mid p) = -1$ | 0 | 2 |
| 4-cycles | $p \equiv 2\ (3)$, $(-1-r \mid p) = +1$ | 2 | 0 |

These are the values of the character of $\mathbf{1} \oplus (\mathrm{sgn}_F \otimes \mathrm{sgn} \otimes \mathrm{std})$ for $\mathrm{Gal}(M/\mathbb{Q}) \cong \mathrm{S}_4 \times \mathbb{Z}/2$, and the observed class densities match the orders of the conjugacy classes. The element (transposition, $\sigma_F$) is complex conjugation and has trace 2, so its eigenvalues are $(1, 1, 1, -1)$. The stabiliser of the anti-invariant class has order 4 in a group of order 48, giving a field of definition of degree 12 — which is the field $L$ of Proposition 4.1. □

**Remark 4.4.** Proposition 4.3 also records where local height contributions can arise: at 3 the stable model is smooth (potentially good reduction), and at 2 the stable model is a pair of supersingular elliptic curves.

## 5. The quadratic Chabauty identity

### 5.1. The imported statements.

**Theorem 5.1** (Balakrishnan–Dogra [BD, Prop. 2], as quoted in [Do])**.** *Let $X/\mathbb{Q}$ be a curve of genus $g$ with Jacobian $J$. The depth-2 Chabauty–Kim set $X(\mathbb{Q}_p)_2$ is finite whenever*

$$\operatorname{rk} J(\mathbb{Q}) \;<\; g - 1 + \operatorname{rk} \mathrm{NS}(J) + \operatorname{rk} \mathrm{NS}(J_{\overline{\mathbb{Q}}})^{c=-1}.$$

Here $2 < 2 - 1 + 1 + 1$, by Proposition 4.3.

**Theorem 5.2** (Balakrishnan–Besser–Bianchi–Müller [BBBM, Thm. 1.5])**.** *Let $X$ be a bielliptic curve of genus 2 over a number field $L$, with quotient maps $\varphi_1, \varphi_2$ to elliptic curves $E_1, E_2$, and let $\chi\colon \mathbb{A}_L^\times / L^\times \to \mathbb{Q}_p$ be a non-trivial continuous idele class character. Then*

$$\rho^\chi(z) = \sum_{\mathfrak{p}\mid p} \Big( h^\chi_{\mathfrak{p}}(\varphi_1(z_{\mathfrak{p}})) - h^\chi_{\mathfrak{p}}(\varphi_2(z_{\mathfrak{p}})) - 2\chi_{\mathfrak{p}}(x(z_{\mathfrak{p}})) \Big) - \sum \alpha\, q_1(\varphi_1(z)) + \sum \beta\, q_2(\varphi_2(z))$$

*maps $X(L)$ into an explicitly computable finite set $T^\chi$.*

We apply Theorem 5.2 to $X_L$, restricted to $X(\mathbb{Q}) \subset X(L)$, for one specific character. The role of the term $\operatorname{rk} \mathrm{NS}(J_{\overline{\mathbb{Q}}})^{c=-1}$ in Theorem 5.1 is the following selection rule, which is what makes the restriction useful: for $z \in X(\mathbb{Q})$ all Galois-conjugate places contribute the same local terms, so the identity survives on $\mathbb{Q}$-points only for pairs (class, character) lying in the same Galois-isotypic component. The cyclotomic character pairs with the trivial component and gives $0 = 0$; the usable classes are the complex-conjugation-anti-invariant ones.

### 5.2. The character.

**Proposition 5.3.** *Let $p = 499$, the smallest prime that splits completely in $M$, so that $L \otimes \mathbb{Q}_p = \mathbb{Q}_p^{12}$ and every endomorphism is $p$-rational. Under the twelve embeddings $\sigma_i\colon L \to \mathbb{Q}_p$ the involutions act on $H^0(X, \Omega^1)$, in the basis $dx/y$, $x\,dx/y$, by*

$$A_i = s_i^{-1} \begin{pmatrix} -1 & -c_i \\ -b_i & 1 \end{pmatrix}.$$

*Because $H^0(\Omega^1)$ is one of the two CM eigenspaces of Frobenius on $H^1_{\mathrm{dR}}$, one has $\mathrm{NS}(J) \otimes \mathbb{Q}_p \cong M_2(\mathbb{Q}_p)$ through this action, and the $A_i$ span $\mathfrak{sl}_2$, the non-trivial 3-dimensional part. Let $R \subset \mathbb{Q}_p^{12}$ be the orthogonal complement of $\ker\big((c_i) \mapsto \sum_i c_i A_i\big)$; then $\dim R = 3$ and $R$ is the isotypic component of $\mathrm{sgn}_F \otimes \mathrm{sgn} \otimes \mathrm{std}$ in the permutation representation on the places above $p$. The six fundamental units of $L$ have $p$-adic logarithms of rank 6, meeting $R$ in a 2-dimensional space. Hence there is, up to scaling, exactly one idele class character $\chi_Z$ of $L$ whose local components at the places above $p$ lie in $R$:*

$$\chi_Z = (c_1, \ldots, c_{12}), \quad c_1 = 1,$$

*extended to all places by $h_L = 1$ via $\chi_w(\pi_w) = \sum_i c_i \log_p \sigma_i(\alpha_w)$ for a generator $\alpha_w$ of $w$. Moreover $\chi_Z$ vanishes at the uniformisers of every prime of $L$ above 2 and above 3.*

*Proof.* `step1_character.gp`: the kernel has dimension 9, so $\dim R = 3$; $R \perp (1, \dots, 1)$; the unit logarithms have rank 6 and $\dim(R \cap \mathrm{span}(\text{unit logs})) = 2$; and the values at the four uniformisers above 2 and 3 are $O(499^{30})$. □

**Remark 5.4.** The vanishing at 2 and 3 is what makes the identity below independent of the reduction types there: no analysis of the bad fibres is required. Whether it has a structural explanation is one of the questions the author would put to an expert.

5.3. **The identity.** Set $D_1 = [P_1 - \iota P_1]$, $D_2 = [P_2 - \iota P_2] \in J(\mathbb{Q})$ and $Q_i = \varphi_1(P_i) \in E_1(L)$, $Q_i' = \varphi_2(P_i) \in E_2(L)$. The 499-adic logarithms of $Q_1, Q_2$ at two places are independent, so $D_1, D_2$ span $J(\mathbb{Q}) \otimes \mathbb{Q}$; the rank is exactly 2 by §7.

**Theorem 5.5.** *For $z \in X(\mathbb{Q})$ write $[z - \iota z] = a_1 D_1 + a_2 D_2$ in $J(\mathbb{Q}) \otimes \mathbb{Q}$ and set $a(z) = (a_1, a_2)^{\mathsf{T}}$, and let*

$$G_{ij} = \langle Q_i, Q_j \rangle_\chi - \langle Q_i', Q_j' \rangle_\chi$$

*be the difference of the $\chi_Z$-height pairings on $E_1(L)$ and $E_2(L)$. Then for every $z \in X(\mathbb{Q})$,*

$$\rho(z) := \sum_{i=1}^{12} c_i \big[\lambda_i(\varphi_1 z) - \lambda_i(\varphi_2 z)\big] \;-\; 2 \sum_{i=1}^{12} c_i \log_p \sigma_i(\mathcal{X}(z)) \;-\; a(z)^{\mathsf{T}} G a(z) \;=\; -\Omega_{397}, \tag{5.1}$$

*where $\lambda_i$ is the canonical Mazur–Tate local height at the $i$-th place above $p$ on the $b$-model of the relevant quotient, and $\Omega_{397}$ is the constant $2\chi'_{w_1}(\pi) - 2\chi'_{w_2}(\pi)$ attached to the two degree-1 primes $w_1, w_2$ of $L$ above 397.*

*Proof.* Since $\varphi_1(z) = a_1 Q_1 + a_2 Q_2 + T$ with $T \in E_1(L)[2]$, and heights kill torsion, $h_\chi(\varphi_1 z) - h_\chi(\varphi_2 z) = a^{\mathsf{T}} G a$. Decompose $h_\chi$ into local terms: at the twelve places above $p$, $c_w \lambda_w(Q) + d_w m_w(Q)$ with $m_w = \max(0, -v_w(x(Q)))$; at a place $w \nmid p$ of good reduction of the minimal model, $\chi'_w(\pi_w) m_w(Q)$; and at the places above 2 and 3, nothing, by Proposition 5.3.

On the natural quotient models $x(\varphi_1 z) = A\mathcal{X}^2$ and $x(\varphi_2 z) = D/\mathcal{X}^2$, so at every place $w$ not above $2, 3, 397$ — where $A, D$ are units and the models are minimal —

$$m_w(\varphi_1 z) - m_w(\varphi_2 z) = -2 v_w(\mathcal{X}(z)), \tag{5.2}$$

and the same holds at $w \mid p$. Summing $\chi'_w(\pi_w) \cdot (-2 v_w(\mathcal{X}(z)))$ over $w \nmid p$, and using that $\chi_Z$ kills the global element $\mathcal{X}(z) \in L^\times$, produces

$$-2 \sum_i c_i \log_p \sigma_i(\mathcal{X}(z)) \;+\; 2 \sum_i d_i v_p(\sigma_i \mathcal{X}(z)),$$

and the $d_i$-terms cancel against the terms $d_i(m_i(\varphi_1 z) - m_i(\varphi_2 z))$ at $p$, by (5.2) again. What remains is the contribution of the two primes above 397, where the natural models are not minimal; this is Lemma 5.6 below, which shows it to be the constant $\Omega_{397}$. □

**Lemma 5.6.** *Let $w$ be one of the two degree-1 primes of $L$ above 397, with $v_w(A) = -6$ at the first and $v_w(D) = -6$ at the second, and let $u_1, u_2$ be the scalings to the minimal models, of valuations $v_w(u_1) = -2$ and $v_w(u_2) = -1$ (discriminant exponents $-24$ and $-12$). Then, computing $m_w$ on the minimal models,*

$$m_w(\varphi_1 z) - m_w(\varphi_2 z) + 2 v_w(\mathcal{X}(z)) = +2$$

*at the first prime, for every 397-adic point $z$, and $-2$ at the second. Hence the correction in (5.1) is the constant $\Omega_{397} = 2\chi'_{w_1}(\pi) - 2\chi'_{w_2}(\pi)$.*

*Proof.* Case analysis on $v := v_w(\mathcal{X}(z))$, using $v_w(x_-) = -1$ and $x_+$ integral. For $v \le 0$: $m_w(\varphi_1) = 2 - 2v$ and $m_w(\varphi_2) = 0$. For $v = 1$: both vanish. For $v \ge 2$: $m_w(\varphi_1) = 0$ and $m_w(\varphi_2) = 2v - 2$. In each case the stated combination equals 2; the value $-2$ at the second prime follows by the conjugation symmetry exchanging the roles of $A$ and $D$. □

**Remark 5.7.** The constant is computed from a generator of the two primes and from $\mathcal{X}(P_1)$. Numerically $\rho(P_k) + \Omega_{397} = O(499^{19})$ at both known points with the sign $-2\log\mathcal{X}$ of (5.1), and this fails with the opposite sign (`step2b.gp`); so the numerics pin the normalisation independently of the derivation.

5.4. **Computing the heights.** Throughout we use the $\sigma$-function description of the local height at $p$, which replaces a double Coleman integral by an evaluation of the Mazur–Tate $\sigma$-function [MT] and is what makes the computation below practical; we follow [Bi] for that description and [BBBM] for its extension to a number field and a non-cyclotomic character.

The heights were computed as follows (`step2b.gp`). Each $E_k \otimes_{\sigma_i} \mathbb{Q}_p$ is replaced by an integer model congruent to it modulo $499^{30}$, so that PARI's `ellpadics2` supplies the $\sigma$-constant; $\sigma$ itself comes from the formal group. With $m = \#E(\mathbb{F}_p) = 514$, the division polynomial $\psi_m(Q)$ is evaluated by the classical recurrences rather than by expanding a polynomial of degree ≈130,000, calibrated by

$$\log\sigma(kQ_0) - k^2\log\sigma(Q_0) = \log\psi_k(Q_0) \qquad (k = 2, 3, 4)$$

to $O(499^{19})$ — which also fixes PARI's division-polynomial parity convention and the $a_1 = a_3 = 0$ requirement of the $\sigma$-identities — and then

$$\lambda(Q) = -\frac{2}{m^2}\big(\log\sigma(mQ) - \log\psi_m(Q)\big).$$

*Validation.* The $\chi_Z$-height is a quadratic form: on both quotients

$$h(Q_1 + Q_2) + h(Q_1 - Q_2) - 2h(Q_1) - 2h(Q_2) = O(499^{19}), \qquad h(2Q_1) = 4h(Q_1).$$

The relative sign between the $p$-adic and away-from-$p$ terms was wrong in a first version, and this test is what caught it.

## 6. Zeros and the Mordell–Weil sieve

**Proposition 6.1.** *$\rho$ extends to a locally analytic function on $X(\mathbb{Q}_p)$. On the disks where some $\sigma_i(\mathcal{X}(z))$ tends to 0 or $\infty$, the singular pair of terms is replaced by its analytic form $2c_i[\log(s/\mathcal{X}) + v(s)]$ with $s$ the formal parameter, as in* [BP, Rem. 2.5]*; the logarithmic singularities of $\lambda_i$ and of $\log\mathcal{X}$ cancel.*

*At $p = 499$, $X(\mathbb{Q}_{499})$ has 528 residue disks and no Weierstrass disk ($C_3$ has no root mod 499). Expanded to $O(T^{14})$ with 14-digit coefficients, $\rho$ has Strassman index 2 on every disk, with coefficient valuations $[\,\cdot, 1, 1, 2, 3, \dots]$ — the quadratic term dominates because every logarithm lies in $499\mathbb{Z}_{499}$ and the inverse log-matrix has valuation $-1$. The $\mathbb{Z}_{499}$-roots of the truncated series number 524 for $B^-$ (262 disks with two roots, 266 with none) and 504 for $C^+$. Recomputation with 20 digits and $O(T^{20})$ reproduces the roots to 8 digits and the counts, the valuations continuing linearly to 18, so the truncation loses nothing. Finally $\rho$ vanishes to $O(499^{13})$ at the known points.*

*Proof.* The expansion and the root-finding are `step3_setup.gp`, `step3_disk.gp` and `step3_run.gp`, aggregated by `step3_aggregate.py`; the higher-precision rerun is `step3_precision_check.gp`. The per-disk output is in `logs/step3_disks.txt`, at 0.74 s per disk, run in four parallel chunks. □

**Proposition 6.2** (The sieve)**.** *Every point of $J(\mathbb{Q}_{499})$ has a 499-integral coefficient vector $a$: all logarithms lie in $499\mathbb{Z}_{499}$, and the log-matrix of $Q_1, Q_2$ has determinant of valuation 2. Hence $[J(\mathbb{Q}) : \langle D_1, D_2\rangle + J(\mathbb{Q})_{\mathrm{tors}}]$ is a 499-unit, and for a degree-1 prime $\mathfrak{q}$ of $L$ above $q$ with $499 \mid \#E_1(\mathbb{F}_\mathfrak{q}) = \#E_2(\mathbb{F}_\mathfrak{q})$, projecting to the 499-parts $\xi\colon E_k(\mathbb{F}_\mathfrak{q}) \to \mathbb{Z}/499$, a rational point $z$ must satisfy*

$$\xi\big(a(z)\cdot(\bar{Q},\bar{Q}')\big) \in \big\{\xi(\varphi_1(w),\varphi_2(w)) : w \in X(\mathbb{F}_q)\big\} \subset (\mathbb{Z}/499)^2,$$

*with no torsion and no index ambiguity, both 2-torsion translates and unit indices vanishing mod 499.*

*Such primes $q$ are: four above 997 (with $\#E = 998$), four above 12973, two above 47903, two above 59879, and twelve above 88339. The first prime above 997 eliminates 518 of the 524 zeros on $B^-$ (and 494 of 504 on $C^+$); the primes above 12973 eliminate all the rest but four. On each curve the four survivors are the four known points: $(1,\pm1)$ and $(-1,\pm15)$ with $a = (\pm1,0),(0,\pm1)$ on $B^-$, and $(2,\pm12)$, $(6/5,\pm172/125)$ likewise on $C^+$.*

*Proof.* `step4_scan.gp`, `step4_sieve.gp`; `logs/step4_sieve.log`. □

6.1. **An independent second prime.** The whole of §5–§6 was repeated at $p = 1459$, the next prime that splits completely in $L$ and at which $C_3$ has no root (`second_prime/make_p1459.py`, which regenerates the pipeline from the $p = 499$ scripts; logs in `logs/p1459/`). The structural facts recur: the character is again unique in its isotypic component and vanishes at the uniformisers above 2 and 3; the $\chi_Z$-heights satisfy the parallelogram law to $O(1459^{19})$; $\rho(P_1) = \rho(P_2) = -\Omega_{397}$ to $O(1459^{19})$ on both curves; and $\#E_k(\mathbb{F}_\mathfrak{p}) = 1462$ at all twelve places, so the reduction is ordinary. Each curve has 1464 residue disks, all of Strassman index 2 except the two disks at infinity (index 1), and $\rho$ has 1546 zeros on $B^-$ and 1426 on $C^+$. The sieve primes are the degree-1 primes of $L$ above $q = 2917$, 26261, 96293, all $\equiv -1 \pmod{1459}$ with $\#E_1(\mathbb{F}_\mathfrak{q}) = \#E_2(\mathbb{F}_\mathfrak{q}) = q + 1$; the first prime above 2917 removes 1538 of the 1546 zeros on $B^-$, and 26261 removes all the rest but four, which are the known points.

On $C^+$ the same primes leave the four known points together with the two zeros of each of the two disks $x \equiv 103 \pmod{1459}$. On those disks the logarithms at places 1 and 2 lose all precision, and the 14-digit run had returned coefficient vectors with no significant digits. Recomputing them with 22 working digits (`logs/p1459/Cplus/disk103.log` and the two follow-up logs) gives the same two zeros to five digits and ordinary coefficient vectors $a \equiv (243,104),(1453,735),(1216,1355),(6,724) \pmod{1459}$, which the prime above 2917 removes. So at $p = 1459$ too, $B^-(\mathbb{Q})$ and $C^+(\mathbb{Q})$ consist of exactly the four known points each.

**Remark 6.3.** This is the one place where the pipeline required a per-disk precision intervention. A systematic per-disk precision check on $a(z)$, and not only on $\rho$, would be the right way to handle it; the present runs check it by hand on the one disk where it was needed.

**Corollary 6.4.** *Granting $\operatorname{rk} J(\mathbb{Q}) = 2$ for both Jacobians (§7),*

$$B^-(\mathbb{Q}) = \{(1,\pm1),\ (-1,\pm15)\}, \qquad C^+(\mathbb{Q}) = \{(2,\pm12),\ (6/5,\pm172/125)\}.$$

## 7. The rank bounds

**Proposition 7.1.** *For $X = B^-$ and for $X = C^+$: $J(\mathbb{Q})[2] = 0$ and $\dim_{\mathbb{F}_2} \mathrm{Sel}^2(J/\mathbb{Q}) = 2$. Hence $\operatorname{rk} J(\mathbb{Q}) \le 2$, so $\operatorname{rk} J(\mathbb{Q}) = 2$ and $Ш(J/\mathbb{Q})[2] = 0$.*

*Proof.* Write $f$ for the sextic ($-C_3$ for $B^-$, $2C_3$ for $C^+$), let $L_6 = \mathbb{Q}[x]/(f)$ and let $\vartheta$ be the image of $x$. In both cases $L_6$ has class number 1, signature $(2,2)$ and Galois group $2 \wr \mathrm{S}_3$ of order 48, and no quadratic subfield (`nfsubfields`).

*(a) Torsion.* $J[2](\overline{\mathbb{Q}})$ is the group of even-cardinality subsets of the roots of $f$ modulo complementation, so a $\mathbb{Q}$-rational 2-torsion point is a Galois-stable such subset, that is, a union of irreducible factors. Since $f$ is irreducible, $J(\mathbb{Q})[2] = 0$.

*(b) The fake Selmer group.* Let $H = \ker\bigl(\mathrm{N}\colon L_6^\times/L_6^{\times 2}\mathbb{Q}^\times \to \mathbb{Q}^\times/\mathbb{Q}^{\times 2}\bigr)$. The map $D = \sum n_P P \mapsto \prod (x_P - \vartheta)^{n_P}$ induces a homomorphism $x - T\colon J(\mathbb{Q}) \to H$, and likewise over every completion, and the fake 2-Selmer group is

$$\mathrm{Sel}^2_{\mathrm{fake}}(J) = \{\delta \in H : \delta_v \in (x-T)(J(\mathbb{Q}_v)) \text{ for all } v\}$$

[PS, §§12–13], [St, §5]; see also [Cr]. It is a quotient of the true Selmer group: there is an exact sequence $\mu_2 \to \mathrm{Sel}^2(J) \to \mathrm{Sel}^2_{\mathrm{fake}}(J) \to 0$ ([SvL, §5], with [PS, Thm. 13.2] for the injectivity of the first map), whence

$$\dim \mathrm{Sel}^2(J) \le \dim \mathrm{Sel}^2_{\mathrm{fake}}(J) + 1.$$

(In fact this is an equality here, the first map being injective because $f$ has no odd-degree factor over $\mathbb{Q}$ and is not a constant times a product of two conjugate cubics over a quadratic field, and the image of $\mu_2$ being the class of $[P - \iota P]$ for a rational point $P$, which is a Selmer element. Only the inequality is used.)

*(c) Local images.* For a finite place $v$, let $m_v$ be the number of irreducible factors of $f$ over $\mathbb{Q}_v$ and $o_v \in \{0,1\}$ indicate whether one has odd degree. Then $\dim J(\mathbb{Q}_v)[2] = m_v - 1 - o_v$, by the same count as in (a), and $\dim J(\mathbb{Q}_v)/2J(\mathbb{Q}_v) = d_v := \dim J(\mathbb{Q}_v)[2] + 2\cdot[v=2]$. The kernel of $x - T$ on $J(\mathbb{Q}_v)/2J(\mathbb{Q}_v)$ is generated by the class $c_v$ of $[P - \iota P]$ for $P \in X(\mathbb{Q}_v)$, which does not depend on $P$ (the difference of two such classes is $2[P - P']$); and $c_v = 0$ exactly when the canonical class is twice a $\mathbb{Q}_v$-rational degree-1 class, i.e. when one of the 16 theta characteristics — the six $[w_i]$ and the ten $[w_i + w_j - w_k]$, one for each partition of the Weierstrass points into two triples — is Galois-invariant, i.e. when $o_v = 1$ or all local factors of $f$ share a quadratic subfield $\mathbb{Q}_v(\sqrt{d})$ (the two orbits of $\mathrm{Gal}(\overline{\mathbb{Q}}_v/\mathbb{Q}_v(\sqrt{d}))$ on the roots of each factor then assembling into a stable partition into triples). Writing $\epsilon_v = 1$ in the remaining case, the image $(x-T)(J(\mathbb{Q}_v))$ has dimension exactly $d_v - \epsilon_v$; the computation generates it by $\mathbb{Q}_v$-points and conjugate pairs of quadratic points and stops when that dimension is reached, so the local conditions imposed are exactly the fake Selmer conditions. At the real place the image is given by sign vectors [BS, §5], and is trivial because $f$ has two real roots. At $p \nmid 2\,\mathrm{disc}(f)$ the image is the unramified subgroup, so $\mathrm{Sel}^2_{\mathrm{fake}}(J) \subseteq H(S) := \{\delta \in H : v_{\mathfrak{p}}(\delta) \text{ even for } \mathfrak{p} \nmid S\}$ for any $S \supseteq \{p \mid 2\,\mathrm{disc}(f)\} = \{2,3\}$; we used $S = \{2,3,17\}$, the prime 17 dividing the leading coefficient, which is harmless.

*(d) Computation* (`sel2/pipeline.gp`; logs in `logs/sel2/descent_Bminus.log` and `descent_Cplus.log`). $L_6$ is taken in a reduced model (`polredbest`; this matters — in the model $c^5 f(y/c)$ PARI's local Hilbert symbols at the primes above 2 are unreliable). $H(S)$ is computed from the $S$-units, the class number being 1, as the norm-square subgroup of $L_6(S,2)$ modulo the image of $\mathbb{Q}(S,2)$, with an assertion that the exponent vectors reconstruct $-1$ and the primes of $S$. Local square classes at $\mathfrak{P} \mid p$ are coordinates with respect to a basis of $L_{6,\mathfrak{P}}^\times/L_{6,\mathfrak{P}}^{\times 2}$ chosen by the rank of the Hilbert pairing matrix (`nfhilbert`), and the local symbols are checked against the product formula on 60 random pairs.

For $B^-$: $\dim H(S) = 2$. At $p = 3$, $f$ is irreducible over $\mathbb{Q}_3$ and 3 is a square in $L_6 \otimes \mathbb{Q}_3$, so $d = 0$, $\epsilon = 0$ and the image is 0. At $p = 17$ the factors have degrees $1, 1, 4$, so $d = 1$, $\epsilon = 0$, and the image has dimension 1, generated by $\mathbb{Q}_{17}$-points. At $p = 2$, $f$ is irreducible over $\mathbb{Q}_2$ and no non-square of $\mathbb{Q}_2$ becomes a square in $L_6 \otimes \mathbb{Q}_2$, so $d = 2$, $\epsilon = 1$, and the image has dimension 1, generated by conjugate pairs of quadratic points (72289 pairs tested; the $\mathbb{Q}_2$-points give nothing). The subgroup of $H(S)$ satisfying the three conditions has dimension

1, so $\dim \mathrm{Sel}^2_{\mathrm{fake}}(J) = 1$ and $\dim \mathrm{Sel}^2(J) \le 2$. Since $\mathrm{rk}\, J(\mathbb{Q}) \ge 2$, by the independent points of §5, and $J(\mathbb{Q})[2] = 0$, we get $\dim \mathrm{Sel}^2(J) = 2$ and $Ш[2] = 0$.

For $C^+$: $\dim H(S) = 2$ and the local data are the same (image of dimension 1 at 2, from 151 quadratic pairs; 0 at 3; 1 at 17), so $\dim \mathrm{Sel}^2_{\mathrm{fake}} = 1$ and $\dim \mathrm{Sel}^2 = 2$. □

7.1. **Validation of the implementation.** The pipeline was run on 24 curves of the genus-2 database of Booker et al. [BSSVY] having rational points and irreducible sextic models (`sel2/validation_curve_*.gp`; scorecard in `logs/sel2/validation_scorecard.txt`). For a curve with a rational point and no 2-torsion one has $\dim \mathrm{Sel}^2(J) = \mathrm{rk}\, J(\mathbb{Q}) + \dim Ш[2]$ with $\dim Ш[2]$ even [PSt], so $\dim \mathrm{Sel}^2 \ge \mathrm{rk}\, J(\mathbb{Q})$ with equality exactly when $Ш[2] = 0$. On all 24 curves the computed $\dim \mathrm{Sel}^2$ equals the database's analytic rank — including curves with four real roots, with class groups $\mathbb{Z}/2$ and $\mathbb{Z}/3$, with a quadratic subfield, and of ranks 1, 2 and 3.

Two curves with leading coefficient 4, a 2-adic square, needed $\mathbb{Q}_2$-points near infinity, with $x$ of denominator up to $2^9$, to complete the 2-adic local image; the pipeline searches that chart when the standard search falls short, and flags any image that remains incomplete, in which case its output is only a lower bound.

Three earlier bugs — the exponent ordering of `bnfissunit`, the Hilbert symbols in the non-reduced model, and an empty local image at a prime with trivial target — had produced $\mathrm{rk}+1$ on ten of the validation curves, which the parity of $\dim Ш[2]$ rules out; that is how they were found.

## 8. Proof of the Theorem

*Proof of Theorem 1.1.* By Lemmas 2.1–2.3, every primitive solution of (1.1) gives a rational point on one of the six curves (2.1), (2.2). By Proposition 3.1, three of them have no rational point; by Proposition 3.2, $A^+(\mathbb{Q}) = \{\infty_\pm, (-1, \pm 3)\}$, which contributes only $(\pm 1, 0, -1)$. By Corollary 6.4 together with Proposition 7.1, $B^-(\mathbb{Q})$ and $C^+(\mathbb{Q})$ consist of the four known points each; transporting them back through Lemma 2.3, the points $(1, \pm 1)$ give $(\pm 1, -1, 1)$, the points $(-1, \pm 15)$ and $(2, \pm 12)$ give an imprimitive solution, and $(6/5, \pm 172/125)$ gives $(\pm 43, -203, 237)$. That exhausts the primitive solutions, and each of the three listed is checked directly to satisfy (1.1). □

The result is subject to the imported statements listed in §1.4: the theorems of Balakrishnan–Dogra and of Balakrishnan–Besser–Bianchi–Müller, the standard theory of Coleman integration used in Proposition 3.2 and in §5, and the theory of $p$-adic heights on elliptic curves over number fields. The $\sigma$-function local heights used here were validated against PARI's canonical heights over $\mathbb{Q}$ in the companion implementation (`supporting/qc_sigma.py`). The $p$-adic computations carry PARI's rigorous precision tracking; the only truncation is the series tail on each residue disk, controlled by the observed linear growth of coefficient valuations and confirmed at higher precision.

## 9. Code and data

All scripts and the logs of every run cited above are public at https://github.com/marchfederico/fermat433: the code under the GNU General Public License, version 3 or later, and this paper and the other written documents under CC BY 4.0. The scripts in `exploration/`, which nothing here depends on, are Denis Simon's and carry his copyright. Everything runs on PARI/GP 2.17.4; the Python scripts need only `sympy`. No Magma or Sage is used. The long runs are the residue-disk expansions (`step3_run.gp` in four chunks, about 10 minutes at $p = 499$ and about 40 minutes at $p = 1459$) and the 2-adic point searches of `sel2/pipeline.gp`, up to about 1.5 hours per curve.

The directory `lean/` contains a Lean 4 / Mathlib formalisation of the reduction. Its main theorem states that, given Lemma 2.1 and the $x$-coordinates of the affine rational points of the six curves (2.1), (2.2) as determined in Propositions 3.1, 3.2 and Corollary 6.4, the primitive solutions of (1.1) are exactly those of Theorem 1.1. These inputs appear as explicit hypotheses of the theorem, not as axioms. What Lean proves is the content of Lemmas 2.2 and 2.3 that the proof uses — that $a$ is odd, that $k = 2, 3$ cannot occur, and the twist set — together with the transport of the points in §8. The formal proof contains no `sorry` and depends only on Lean's three standard axioms; the repository's continuous integration rebuilds it and checks this on every change.

TABLE 1. Where each statement is established.

| Statement | Scripts | Logs |
|---|---|---|
| Lemmas 2.1–2.2 | `descent_explore.py`, `diagonal.py` | — |
| Lemma 2.3 | `etale.py`, `twistset.py` | — |
| Lemmas 2.2–2.3 and §8, formally | `lean/Fermat433/` | — |
| Prop. 3.1 | `sel2/certify.gp`, `sel2/fakeselmerset.gp` | `logs/sel2/certify.log` |
| Prop. 3.2 | `aplus/bi_Aplus.gp`, `aplus/chab5_Aplus.gp`, `aplus/sieve5_Aplus.gp`, `aplus/tors12.gp` | `logs/aplus/` |
| Prop. 4.1 | `invol.gp`, `invol4.gp`, `invol8.gp`, `step2a.gp` | — |
| Prop. 4.3 | `nschar2.gp`, `splitF.gp` | — |
| Prop. 5.3 | `step1_character.gp` | `logs/step1_data.txt` |
| Thm. 5.5 | `step2a.gp`, `step2b.gp` | `logs/step2b_data.txt` |
| Prop. 6.1 | `step3_*.gp`, `step3_aggregate.py` | `logs/step3_disks.txt` |
| Prop. 6.2 | `step4_scan.gp`, `step4_sieve.gp` | `logs/step4_sieve.log` |
| Second prime | `second_prime/make_p1459.py` | `logs/p1459/` |
| $C^+$ throughout §§4–6 | `twist/` | `logs/twist/` |
| Prop. 7.1 | `sel2/pipeline.gp` | `logs/sel2/` |
| Validation | `sel2/validation_curve_*.gp` | `logs/sel2/validation_scorecard.txt` |

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